Stage-Based Nutrition Recommendation for Lactating Mothers Using an Interpretable Machine Learning Model

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Abstract

Adequate maternal nutrition during lactation is essential for supporting maternal health and meeting the changing nutritional demands of breastfeeding. However, lactating women remain underserved by personalised digital nutrition systems, particularly those that account for lactation stage while providing transparent and practical dietary recommendations. Existing approaches often address nutritional requirement estimation, model interpretability, and dietary optimisation separately, limiting their integration into accessible maternal nutrition support systems. This study addresses this gap by developing a stage-based nutrition recommendation framework that combines interpretable machine learning with mathematical optimisation. Demographic and Health Survey (DHS) data from 12,176 breastfeeding women across five African countries were used to train separate Generalised Additive Models (GAMs) for estimating personalised daily energy and protein requirements across early, mid, and late lactation. Reference-based micronutrient requirements were incorporated alongside the GAM predictions into a Mixed-Integer Linear Programming (MILP) model, which generated daily meal plans using USDA food composition data and curated recipes while considering dietary preferences, allergies, and food exclusions. On a held-out test set of 2,436 observations, the energy GAM achieved an R^2 of 0.9722 and a Mean Absolute Error (MAE) of 26.86 kcal/day, while the protein GAM achieved an R^2 of 1.0000 and an MAE of 0 g/day. Controlled lactation-stage experiments produced energy targets of 2,265, 1,977, and 2,328 kcal/day for early, mid, and late lactation, respectively, with corresponding changes in meal selections and portion quantities. Functional testing demonstrated successful meal generation across unrestricted, vegetarian, vegan, and dietary-restricted scenarios, with validated plans meeting 98–102% of predicted energy requirements, 95–105% of predicted protein requirements, and all active hard micronutrient constraints. However, the GAM performance reflects agreement with equation-derived nutritional targets rather than independent clinical validation, while vitamin B12 and D remained challenging in some restrictive diets. Overall, the findings demonstrate the computational feasibility of integrating interpretable machine learning and dietary optimisation into a personalised, stage-aware maternal nutrition recommendation system.

Keywords: Interpretable Machine Learning, Generalised Additive Models (GAMs), Maternal Nutrition, Lactation Stages, Mixed-Integer Linear Programming, Personalised Nutrition, Web-Based Nutrition Recommendation System, Dietary Optimisation.

1. Introduction

The postpartum period represents a critical window for maternal recovery and infant development. However, a significant global policy deficit exists: half of all countries lack Food-Based Dietary Guidelines (FBDGs), and only 15% offer advice tailored specifically for lactation. In digital health, integrating artificial intelligence is currently stalled by a socio-technical impasse: conventional deep neural networks act as opaque black boxes, leaving clinicians and mothers unable to audit the logic behind specific recommendations. Because medical reliability requires transparency, uninterpretable predictions impede clinical trust and prevent informed, accountable decision-making [1–3].

Fluctuating metabolic demands during human milk maturation create non-linear nutritional shifts that can deplete maternal reserves, increasing the risk of postpartum anaemia and osteoporosis. To model these complex shifts transparently, this study implements a Generalised Additive Model (GAM), an intrinsically interpretable glass-box architecture. GAMs represent non-linear relationships across maternal attributes (age, weight, stage) while providing additive smooth functions that reveal the exact mathematical contribution of each feature to the final recommendation [1, 4].

This project aims to bridge the faithfulness-interpretability gap by ensuring that recommendations are both human-readable and mathematically faithful to internal model logic. Implemented as a user-friendly web application, the system delivers actionable recourse, offering mothers clear explanations of how specific dietary adjustments directly mitigate health risks across diverse socioeconomic settings [3].

The remainder of this paper is structured as follows: Section 2 reviews related work in maternal nutrition, interpretable machine learning, and dietary recommendation systems. Section 3 presents system design and architecture, including data preparation, GAM development, micronutrient requirements, and MILP formulation. Section 4 describes the implementation of the integrated web-based application and evaluation procedures. Section 5 presents the experimental results and system evaluation, while Section 6 discusses the findings, limitations, and comparison with existing research. Finally, Section 7 concludes the study and outlines directions for future research.

2. Literature Review

2.1 Maternal Nutrition and Lactation Stages

Lactation is a nutritionally demanding period in which maternal dietary intake must support both maternal health and the production of breast milk. Nutritional demands may vary throughout the lactation period as maternal energy expenditure and breastfeeding patterns change. Consequently, dietary guidance for lactating women should account for the stage of lactation rather than treating the entire breastfeeding period as nutritionally uniform. Despite this need, lactation-specific dietary guidance remains limited internationally, with many countries lacking Food-Based Dietary Guidelines (FBDGs) that specifically address breastfeeding women [2, 5].

Stage-aware nutritional guidance is therefore relevant when developing personalised recommendation systems for lactating women. In the present study, lactation is represented as early (0–6 months), mid (7–12 months), and late ($>$ 12 months), enabling lactation stage to contribute to the estimation of maternal nutritional requirements. Micronutrient requirements are incorporated separately using established dietary reference values for lactating women, providing nutritional targets for the subsequent food recommendation process [6, 7].

2.2 Personalised Nutrition and Nutrition Recommendation Systems

Personalised nutrition aims to adapt dietary guidance to characteristics and requirements of an individual rather than providing uniform recommendations to an entire population. Digital nutrition recommendation systems can facilitate this process by combining user information, nutritional requirements, food composition data, and computational methods to generate dietary guidance [1, 8]. For lactating women, this approach is particularly relevant because maternal characteristics and lactation stage can affect nutritional requirements [6], while practical factors such as dietary preferences and food restrictions influence which recommendations are appropriate.

Ramos-Garcia et al. [1] demonstrated this concept through the NUTRISHIELD study, which developed a personalised nutrition platform for lactating mothers and their infants during the first six months of life. The platform integrated clinical records, dietary assessments and laboratory analyses, including information relating to human milk composition and microbiota, to provide personalised reports through a mobile application. This demonstrates the potential of digital systems to translate diverse health and nutrition information into personalised guidance. However, the extensive biological and clinical information required by such systems may not always be readily available, motivating investigation into recommendation approaches that can operate using more routinely obtainable maternal information.

2.3 Interpretable Machine Learning in Nutrition

Machine learning is increasingly used in health and nutritional applications, but predictive performance alone may be insufficient when model outputs contribute to health-related guidance. Complex black-box models can make it difficult to determine how individual characteristics contribute to a prediction. Interpretable Machine Learning (IML) addresses this limitation by providing mechanisms through which the relationship between model inputs and outputs can be examined [3]. This is particularly relevant in health-related applications where transparency can support the understanding

and evaluation of model behaviour.

Recent research has specifically investigated interpretability within nutritional applications. Revesai and Kogeda [9] proposed knowledge-guided neural networks for transparent micronutrient deficiency detection, incorporating nutritional domain knowledge into the modelling process. In related work, Revesai and Kogeda [10] compared interpretable deep-learning models for nutrient analysis in vulnerable populations, further demonstrating the relevance of interpretability when machine learning is applied to nutritional assessment. These studies indicate that transparency is an important consideration alongside predictive performance in computational nutrition.

The present study shares this emphasis on interpretable nutritional modelling but adopts an intrinsically interpretable approach through Generalised Additive Models rather than a neural-network architecture. This allows relationships between maternal characteristics and predicted nutritional requirements to be inspected directly through the structure of the model.

2.4 Generalised Additive Models

Generalised Additive Models (GAMs) provide a flexible modelling framework in which the predicted outcome is represented through the additive contribution of predictor functions. Unlike conventional linear models, GAMs can represent non-linear relationships while retaining a structure that allows the influence of individual predictors to be examined [11]. This combination of flexibility and transparency makes GAMs suitable for applications in which understanding model behaviour is important [12].

Awe et al. [12] investigated GAM-based approaches for obesity risk prediction in a public-health context. Their work demonstrated how additive modelling can capture complex relationships between health-related variables while retaining interpretability. Although their application focused on obesity and associated health indicators rather than maternal nutrition, it provides evidence for the applicability of GAM-based modelling in health-related prediction.

In this study, separate GAMs are used to estimate daily energy and protein requirements from maternal and lactation-related characteristics. Their additive structure enables the effects of variables such as age, weight, height and lactation stage to be examined, providing an interpretable modelling stage before the estimated requirements are passed to the dietary optimisation component.

2.5 Mixed-Integer Linear Programming for Dietary Optimisation

Mathematical optimisation provides a means of translating nutritional requirements into combinations of foods that satisfy defined constraints. Linear Programming (LP) has been applied to dietary planning by representing food quantities as decision variables and nutritional requirements as mathematical constraints [8]. Where the optimisation problem includes discrete or binary decisions, such as whether a particular food or meal component is selected, the problem can be formulated as Mixed-Integer Linear Programming (MILP).

Yu et al. [8] applied Linear Programming to translate nutrient recommendations into optimised diets for Chinese urban lactating women. Their approach developed stage-specific Food-Based Dietary Guidelines while considering nutritional requirements, cost and local dietary practices. Their findings also highlighted the difficulty of satisfying certain micronutrient requirements through food combinations. This demonstrates the importance of explicitly incorporating nutrient constraints into dietary optimisation.

The present study extends this general optimisation concept by combining personalised GAM-derived energy and protein requirements with reference-based micronutrient requirements within a MILP recommendation model. The optimisation process additionally considers practical constraints including portion limits, meal structure, dietary type, allergies, cuisine preferences and user-specified food exclusions. This enables nutritional requirements to be translated into a structured daily meal plan rather than being presented only as numerical targets.

2.6 Related Work

The reviewed studies address complementary aspects of the problem investigated in this research. NUTRISHIELD [1] demonstrates the potential of personalised digital nutrition guidance for lactating mothers but relies on extensive clinical, dietary and biological information. In contrast, the present study uses routinely available maternal characteristics and lactation stage to estimate personalised

energy and protein requirements and integrates these estimates into an interpretable recommendation process.

The dietary optimisation approach of Yu et al. [8] is closely related to the food recommendation component of this research. Both approaches use mathematical optimisation to translate nutritional requirements into food-based recommendations for lactating women. However, Yu et al. focused on optimised dietary guidelines, whereas the present framework integrates optimisation with GAM-derived personalised requirements, reference-based micronutrient targets, individual dietary restrictions and a web-based recommendation interface.

The GAM research of Awe et al. [12] and the interpretable nutritional modelling investigated by Revesai and Kogeda [9, 10] provide complementary support for the interpretability component of the proposed framework. These studies demonstrate the relevance of transparent modelling in health and nutrition applications. The present study applies this principle specifically to stage-based maternal nutrition and connects the interpretable model outputs to a subsequent MILP food recommendation process.

2.7 Research Gap

Existing research has explored personalised digital nutrition support for lactating mothers [1], mathematical optimisation of lactation diets [8], GAM-based interpretable modelling in healthcare [12], and interpretable machine-learning approaches for nutritional analysis [9, 10]. However, these components have largely been investigated separately. A gap remains in integrating stage-aware maternal information, intrinsically interpretable estimation of personalised energy and protein requirements, reference-based micronutrient requirements, and mathematical dietary optimisation within a single accessible recommendation framework.

This study addresses this integration gap by using GAMs to estimate personalised energy and protein requirements from maternal characteristics and lactation stage, incorporating established micronutrient reference values, and applying MILP to translate these nutritional targets into practical daily food recommendations. The resulting framework additionally accounts for dietary preferences and restrictions and presents the recommendations through a web-based interface, connecting interpretable nutritional requirement estimation with practical food-level guidance.

3. System Design and Architecture

3.1 Research Design

This study used a quantitative computational research design to develop and evaluate a stage-based nutrition recommendation system for lactating women. The proposed framework integrates Generalised Additive Models (GAMs) for estimating personalised energy and protein requirements, reference-based micronutrient requirements, and Mixed-Integer Linear Programming (MILP) for generating nutritionally constrained daily meal recommendations. The system architecture was designed to support the processing of maternal characteristics, estimation of nutritional requirements, dietary preference filtering, food optimisation, and recommendation validation. The following subsections describe the data sources, modelling approaches, nutritional constraints, and optimisation framework, while Section 4 presents the implementation of these components within the web-based application.

3.2 System Architecture

Figure 1 illustrates the integrated workflow. The web interface collects maternal characteristics and dietary preferences. Trained GAMs estimate daily energy and protein requirements, while a separate module determines micronutrient planning targets. Food and recipe candidates are filtered for dietary type, allergies, cuisine preferences and excluded ingredients before the MILP selects meals and portions. Final validation checks nutritional and user-defined constraints before the daily recommendation is displayed. This separation distinguishes model-derived targets, reference requirements and food optimisation.

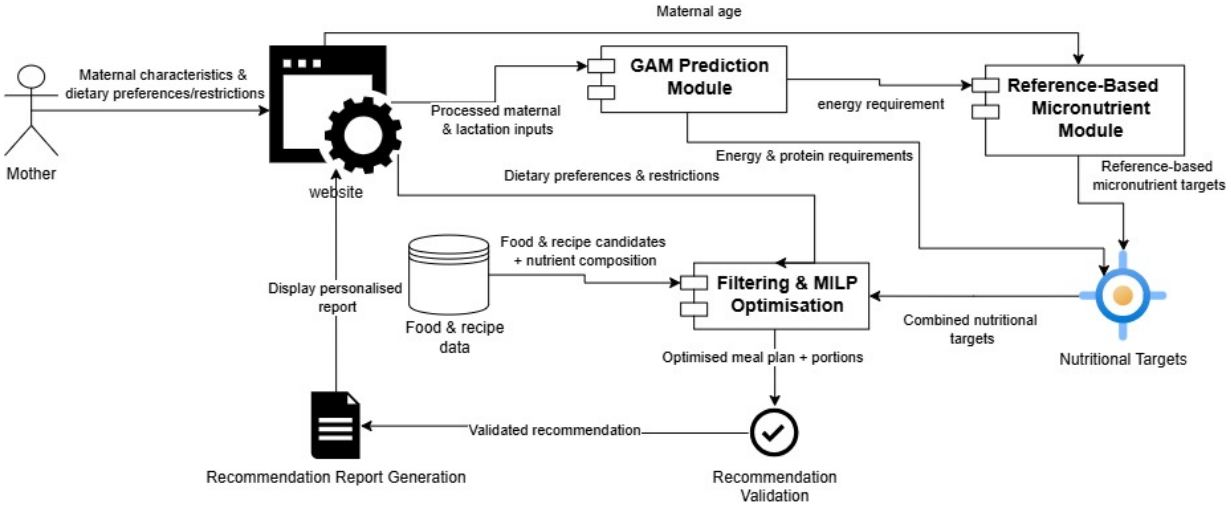


Figure 1: System architecture of the stage-based nutrition recommendation framework.

3.3 Data Sources and Dataset Preparation

3.3.1 DHS Dataset and Participant Selection

This study utilised publicly available Demographic and Health Survey (DHS) Individual and Birth Recode datasets from Burundi (2016) [13], Ethiopia (2024) [14], Kenya (2022) [15], Nigeria (2024) [16], and South Africa (2016) [17]. Linking maternal records with data for the youngest child yielded an initial pool of 118,384 records. Selection was restricted to currently breastfeeding women; following duplicate removal and filtering for complete anthropometric and child-age variables, the final validated dataset comprised 12,176 observations.

3.3.2 Feature Engineering and Lactation Stages

Eight primary input features were selected based on maternal physiological relevance and population context: age, parity, weight, height, pregnancy status, child age, lactation stage, and country. Child age in months was used to classify lactation stage into Early (0–6 months), Mid (7–12 months), and Late (>12 months). One-hot encoding of categorical variables (Country and Lactation Stage) expanded the 8 conceptual inputs into 14 model features for Generalised Additive Model (GAM) training.

3.3.3 Construction of Energy and Protein Targets

Due to the absence of observed individual-level nutritional outcomes, continuous targets for daily protein and energy were deterministically generated using established nutritional equations: * Protein Target: Calculated as 1.1 g/kg body weight/day. * Energy Target: Derived via the Estimated Energy Requirement (EER) framework, incorporating maternal attributes, stage-specific surpluses (+330 kcal/day for Early; +400 kcal/day for Mid), and an 1,800 kcal/day floor.

All generated targets were validated against independent recalculations prior to model training, confirming implementation accuracy.

3.4 Generalised Additive Model Development

3.4.1 GAM Model Selection and Architecture

Generalized Additive Models (GAMs) were selected to represent complex, non-linear relationships while maintaining an intrinsically interpretable additive structure. Two separate LinearGAM models were developed for daily energy (kcal/day) and daily protein (g/day) targets, using the 14 preprocessed features derived from eight primary maternal and population attributes.

3.4.2 Training and Evaluation

The dataset of 12,176 observations was partitioned using an 80/20 train-test split (9,740 training; 2,436 testing; random state = 42). Model performance on the held-out test set was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2. Because the target variables were generated from established equations, these evaluation metrics quantify how faithfully the GAMs

reproduced the underlying target-generation logic, rather than asserting independent clinical validity.

3.4.3 Model Interpretability Analysis

Partial-effect plots were generated across all numerical and categorical predictors to visualize individual feature contributions. This analysis audited the models' behavior to ensure alignment with the mathematical structure of the deterministic equations, confirming algorithmic fidelity without inferring unobserved physiological causality.

3.5 Reference-Based Micronutrient Requirements

Micronutrient targets were obtained from dietary reference values for lactating women published by the Health Council of the Netherlands [18], rather than predicted by the GAMs. The implemented targets covered vitamins A, B1, B2, B6, B12, C and D, folate, calcium, iron, magnesium, selenium and zinc. Calcium requirements varied by maternal age, and vitamin B1 depended on the personalised energy requirement.

Applicable nutrients were enforced as MILP minimums only when at least three eligible foods had positive recorded values for the nutrient. Vitamin D remained a displayed soft planning target; vitamin B12 was report-only for vegan plans; and iodine was excluded because of insufficient food-composition coverage. These reference values serve as planning targets, not assessments of individual nutrient status. Appendix D provides the values and implementation conditions.

3.6 Food Data and Recommendation Catalogue

3.6.1 Food Composition Dataset and Preprocessing

Food composition data used by the recommendation component were obtained from the USDA FoodData Central database, using records from the FNDDS and Foundation food datasets [19]. The raw food data were preprocessed to retain foods with sufficient nutritional information for use within the Mixed-Integer Linear Programming (MILP) recommendation model. Foods unsuitable for the intended recommendation context, including infant foods, alcoholic products, and records lacking essential energy or protein information, were removed. Additional processing was performed to standardise food descriptions, nutrient values, serving information, meal suitability, dietary classification, and other attributes required by the recommendation system.

Following preprocessing and eligibility filtering, foods were classified according to their suitability for different meal roles, including breakfast, lunch and supper, snacks, desserts, and supporting foods. Nutrient composition values were standardised on a per-100 g basis, allowing the MILP model to calculate nutrient contributions according to the portion quantity selected during optimisation.

The resulting food catalogue provided information on energy, protein, macronutrients, and the vitamins and minerals required by the recommendation framework. Iodine was not included as an optimisation constraint because coverage within the available food composition data was insufficient for reliable recommendation planning.

3.6.2 Curated Recipe Dataset

To improve the practicality, variety, and cultural relevance of the generated meal plans, the USDA-derived food catalogue was supplemented with a manually curated dataset containing 211 recipes and 989 ingredient-level records. The recipes represented eight cuisine categories: African, Asian, European, Indian, Latin American, Mediterranean, Middle Eastern, and North American. Seven cuisine categories contained 26 recipes each, while the Asian category contained 29 recipes.

The final curated catalogue comprised 107 vegan-candidate, 56 vegetarian, and 48 omnivore recipes. Recipes were also classified according to meal type, comprising 48 breakfast, 50 lunch, 32 snack, 49 supper, and 32 dessert recipes. These classifications enabled recipes to be filtered according to the user's selected dietary type and incorporated into the appropriate meal position.

Each recipe was linked to its constituent foods in the USDA-derived catalogue using ingredient identifiers. Nutritional composition was calculated from the associated USDA ingredient records and their recorded gram quantities. Nutrient values were standardised per 100 g for optimisation, while serving-level information was retained for presentation and review.

A separate non-nutrient seasoning field was used where seasoning information was necessary to

preserve the identity and description of a prepared dish. These seasonings were included for descriptive and preparation purposes only and were not included in nutrient calculations because nutritional calculations were restricted to ingredients linked to the USDA-derived food catalogue.

3.6.3 Nutrient Composition

For both individual foods and curated recipes, the recommendation catalogue contained the nutritional attributes required by the optimisation model. These included energy and protein together with relevant macronutrient and micronutrient composition values. Nutrient values were represented on a standardised per-100 g basis so that the nutritional contribution of each selected food or recipe could be calculated according to its assigned portion quantity.

For curated recipes, nutrient composition was derived from the quantities and nutrient profiles of the USDA-linked ingredients contained in each recipe. This provided a consistent nutritional basis for comparing individual foods and prepared dishes within the MILP recommendation model.

3.6.4 Dietary, Allergen, and Cuisine Classification

Food and recipe records were assigned classification information required for user-specific filtering. These attributes included dietary suitability, meal type, cuisine category, and relevant allergen information. Dietary classifications enabled the candidate catalogue to be restricted according to the user's selected dietary type, including vegetarian and vegan options, while cuisine classifications supported optional cuisine-based filtering.

Allergen and user-specified food-exclusion filtering was applied before optimisation so that incompatible candidates were removed from the eligible food pool. For curated recipes, filtering considered the recipe name, ingredient names, and ingredient identifiers rather than relying solely on the displayed recipe name. This enabled restrictions to be applied to ingredients contained within a prepared dish and reduced the likelihood of a restricted ingredient remaining hidden within an otherwise eligible recipe.

3.7 Mixed-Integer Linear Programming Recommendation Model

The food recommendation component used Mixed-Integer Linear Programming (MILP) to translate the personalised nutritional requirements generated by the preceding components into a practical daily meal plan. The personalised energy and protein requirements estimated by the trained GAMs were combined with the reference-based micronutrient requirements described in Section 3.5. The optimisation model then selected foods and curated recipes from the eligible recommendation catalogue described in Section 3.6 while accounting for nutritional adequacy, portion limits, meal structure, dietary restrictions, and food variety.

3.7.1 Decision Variables and Objective

For each eligible food or curated recipe i, the optimisation model used a continuous decision variable x_i to represent the quantity of the item selected in grams and a binary decision variable y_i to indicate whether the item was included in the daily plan. The relationship between these variables allowed the model to determine both which foods were selected and the quantity assigned to each selected item.

The optimisation problem can be represented generally as:

$$\min Z(x, y) \tag{1}$$

subject to:

$$0.98E_{\mathrm{GAM}} \leq E(x) \leq 1.02E_{\mathrm{GAM}}, \tag{2}$$

$$0.95P_{\mathrm{GAM}} \leq P(x) \leq 1.05P_{\mathrm{GAM}}, \tag{3}$$

$$N_n(x) \geq R_n, \qquad n \in H, \tag{4}$$

$$L_i y_i \leq x_i \leq U_i y_i, \tag{5}$$

$$x_i \geq 0, \tag{6}$$

$$y_i \in \{0, 1\}. \tag{7}$$

Here, E_{GAM} and P_{GAM} represent the personalised daily energy and protein requirements estimated by the GAMs; $E(x)$ and $P(x)$ represent the energy and protein supplied by the selected foods; $N_n(x)$ represents the total intake of micronutrient n; R_n represents its applicable reference requirement; H represents the set of active hard micronutrient constraints; and L_i and U_i represent the permitted lower and upper portion bounds for food or recipe i.

The objective function $Z(x, y)$ guided the optimiser towards nutritionally adequate and practical food combinations while incorporating the system's soft food-selection preferences. Hard nutritional and structural requirements were enforced separately through the model constraints.

3.7.2 Nutritional Constraints

Energy intake was constrained to remain within 98–102% of the personalised energy requirement estimated by the energy GAM. Protein intake was constrained to remain within 95–105% of the personalised protein requirement estimated by the protein GAM. These ranges provided limited flexibility for constructing feasible food combinations while maintaining close agreement with the personalised nutritional targets.

Micronutrient requirements were incorporated according to the reference-based framework described in Section 3.5. Where sufficient nutrient-composition data were available within the eligible candidate-food pool, applicable micronutrients were enforced as minimum requirements. A micronutrient was activated as a hard constraint only when sufficient usable composition data were available among the eligible candidates.

Vitamin D was retained as a nutritional planning target but was not enforced as a hard feasibility constraint. For vegan recommendations, vitamin B12 was similarly retained as a reported planning target rather than a hard constraint. Iodine was not included as an optimisation constraint because coverage in the available food composition data was insufficient for reliable recommendation planning.

This treatment allowed the optimisation model to enforce nutritional requirements supported by the available food data without making otherwise appropriate meal plans infeasible solely because of limitations in nutrient-composition coverage.

3.7.3 Meal Structure and Portion Constraints

The optimisation model was designed to construct a complete daily meal plan containing breakfast, lunch, snack, supper, and an optional dessert. Candidate foods and curated recipes were assigned to appropriate meal roles so that an item could only be selected for a meal position for which it had been classified as suitable.

Portion constraints were applied to maintain practical serving quantities and prevent the optimiser from satisfying nutritional requirements through unrealistic quantities of individual foods. Each candidate was therefore associated with permitted lower and upper portion bounds. Curated main dishes and supporting foods were treated according to their applicable portion rules, with main dishes permitted within their defined serving limits and supporting foods restricted according to their assigned food subtype.

Additional selection and variety constraints were used to reduce repetitive or impractical combinations. These included restrictions intended to prevent unnecessary duplication of similar supporting foods and to maintain appropriate representation of foods such as vegetables within the daily plan.

3.7.4 Dietary Preferences, Allergies and Food Exclusions

Before optimisation, the candidate-food pool was filtered according to the user's selected dietary type, allergies, cuisine preference, and user-specified food exclusions. Dietary filtering restricted candidates to foods and recipes compatible with the selected omnivore, vegetarian, or vegan dietary pattern, while allergen filtering removed candidates containing ingredients associated with the selected allergies.

Cuisine preferences were applied to the curated recipe catalogue so that recommendations could reflect the user's selected cuisine where applicable. User-specified food exclusions were applied to both individual foods and curated recipes. For curated recipes, filtering considered recipe names, ingredient names, and ingredient identifiers, enabling a recipe to be excluded when a restricted food occurred as an ingredient even if it was not evident from the displayed recipe name.

Filtering was performed before optimisation so that incompatible foods and recipes were not available for selection by the MILP model.

3.7.5 Recommendation Validation

Following optimisation, the generated meal plan underwent a final validation stage before being presented to the user. The validation procedure recalculated the nutritional contribution of the selected foods and portions and verified that total energy and protein remained within their permitted ranges and that the applicable active hard micronutrient requirements were satisfied.

The generated plan was also checked against the user's dietary type, allergies, cuisine-related restrictions where applicable, and user-specified food exclusions. Meal structure and portion quantities were additionally checked to ensure that the recommendation complied with the structural rules applied by the optimisation framework.

Only recommendations that passed the applicable validation checks were returned to the user. Where no combination of eligible foods and recipes could satisfy the active nutritional, structural, and user-defined constraints, the system reported that a feasible recommendation could not be generated rather than presenting a plan that violated the specified requirements.

4. System Implementation

The proposed nutrition recommendation framework was implemented as a web-based application integrating the trained GAM models, reference-based micronutrient requirements, food and recipe catalogues, MILP optimisation, recommendation validation, and report generation. The application enables users to obtain personalised nutritional targets and meal recommendations without directly interacting with the underlying models.

4.1 Implementation Environment

The front-end interface was developed using HTML, CSS, and JavaScript to collect user information, communicate with the Flask backend, and display nutrition recommendations. The application was executed locally during development and evaluation.

4.2 Backend and Model Integration

The Flask backend coordinated the recommendation pipeline. User inputs were validated and processed by the trained energy and protein GAMs to generate personalised nutritional requirements. Reference-based micronutrient targets were then determined according to Section 3.5.

The food and recipe catalogue was filtered according to dietary type, allergies, cuisine preferences, and food exclusions. The resulting eligible foods and nutritional targets were supplied to the MILP optimiser, which selected suitable foods and portion quantities while enforcing the nutritional and structural constraints described in Section 3.7. Generated recommendations underwent final validation before being returned to the interface.

4.3 Web-Based User Interface

The web interface collects maternal characteristics, including age, weight, height, parity, pregnancy status, country, and lactation-stage information. Users can also specify dietary type, allergies, cuisine

preferences, and excluded foods.

JavaScript was used for client-side functionality and input validation, while Flask processed submitted requests. The interface displays personalised recommendations or an appropriate message when no feasible plan can be generated. Nutritional information is presented in an accessible format without requiring users to understand the underlying GAM or MILP models.

4.4 Recommendation Report Generation

Following successful optimisation and validation, the system generates a structured daily nutrition report containing recommended meals, portion quantities, and practical household serving information where available.

The report presents personalised energy and protein requirements alongside vitamin and mineral contributions relative to their reference-based targets. Additional nutrients, including fat, carbohydrates, fibre, and sodium, are displayed for informational purposes rather than as primary optimisation targets.

Simplified explanations relate the recommendations to maternal characteristics and lactation stage. Micronutrients not enforced as hard constraints are identified as requiring attention when their planning targets are unmet. If no feasible meal plan satisfies the active constraints, the system displays an appropriate message instead of generating an invalid recommendation.

4.5 Evaluation and Testing

The developed system was evaluated at both the machine-learning and recommendation-system levels. The GAM component was evaluated using the held-out 20% testing subset described in Section 3.4. Predictive performance was assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2). Model interpretability was evaluated using GAM effect plots to examine how numerical and categorical predictors influenced the estimated energy and protein requirements.

The integrated recommendation system was evaluated using a series of functional test scenarios covering unrestricted, vegetarian and vegan diets, allergies, cuisine preferences, user-specified food exclusions, and combinations of multiple restrictions. Generated recommendations were checked for compliance with the permitted energy and protein ranges, active hard micronutrient constraints, dietary and allergen restrictions, food exclusions, meal structure, and portion requirements. Restrictive scenarios were additionally used to examine system behaviour when the eligible candidate-food pool was substantially reduced.

A controlled lactation-stage experiment was also conducted to determine whether changes in lactation stage propagated through the complete recommendation pipeline. Maternal characteristics and food preferences were held constant while child age was varied to represent Early (0–6 months), Mid (7–12 months), and Late ($>$ 12 months) lactation. The resulting energy and protein requirements, meal selections, portion quantities, and nutritional adequacy were compared across the three conditions.

A second controlled experiment was conducted to examine the effect of maternal weight on the integrated recommendation pipeline. Maternal weight was varied across 55 kg, 65 kg, and 75 kg while the remaining maternal characteristics, lactation-related information, and food preferences were held constant. The resulting energy and protein requirements and generated meal recommendations were compared across the three conditions. This experiment was used to assess whether changes in maternal weight were reflected in the GAM-derived nutritional requirements and subsequently propagated to the MILP food-recommendation component.

5. Results

5.1 Dataset Characteristics and Distribution

The final validated dataset comprised 12,176 observations (majority aged 18–49) spanning Late ($n = 4,920$), Early ($n = 3,910$), and Mid ($n = 3,346$) lactation stages across eight input variables. Continuous targets showed substantial variation, yielding 9,388 unique energy targets (up to 3,410.66 kcal/day). An Estimated Energy Requirement (EER) sanity check confirmed 100% implementation accuracy of the 1,800 kcal floor and stage-specific surpluses.

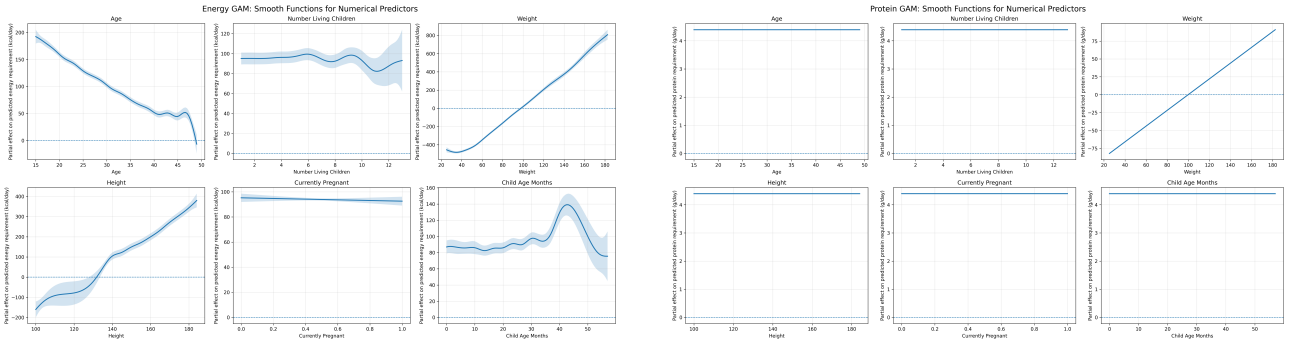
5.2 GAM Performance Outcomes

- **Protein GAM:** Achieved an R^2 of 1.0 (MAE = 0), reflecting exact reproduction of the deterministic weight-based formula.
- **Energy GAM:** Achieved an R^2 of 0.9722 and an MAE of 26.86 kcal/day, demonstrating high fidelity to the multi-variable EER logic.

5.3 Model Interpretability

5.3.1 Numerical Predictor Effects

Energy GAM predictions were strongly positively driven by weight and height and negatively by age, aligning directly with the EER equation. Protein GAM predictions depended almost exclusively on maternal weight.

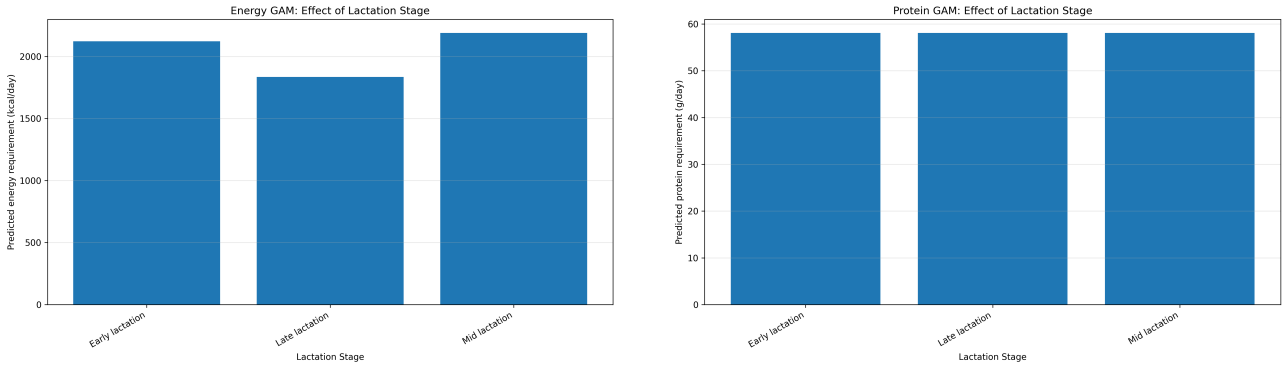


(a) Numerical predictor effects for the Energy GAM. **(b)** Numerical predictor effects for the Protein GAM.

Figure 2: Overview of Numerical Effects.

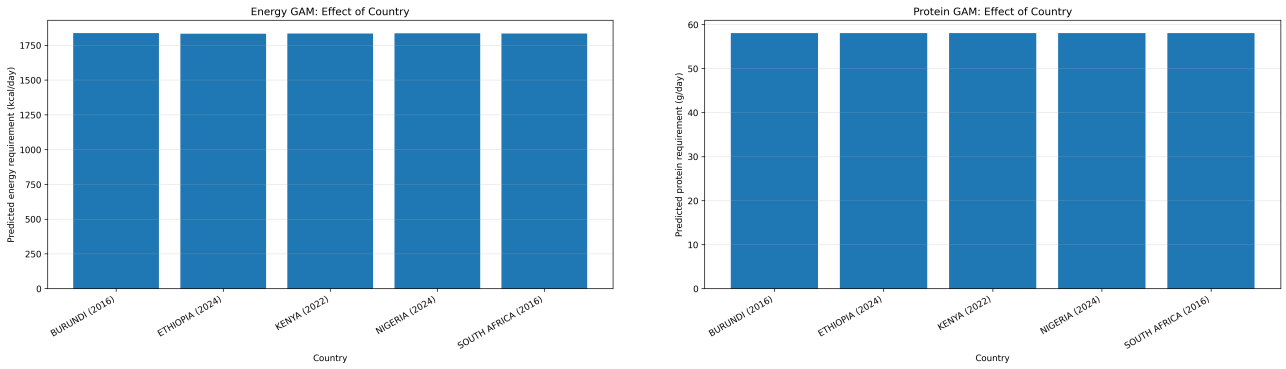
5.3.2 Categorical Effects (Lactation Stage and Country)

Energy GAM isolated stage effects peaked in Mid-lactation (+400 kcal) and dropped lowest in Late-lactation (+0 kcal), whereas Protein GAM showed zero stage variation. Both models exhibited minimal variation across the five countries (Burundi, Ethiopia, Kenya, Nigeria, South Africa), as national variables were not encoded in target equations.



(a) Lactation-stage effects for the Energy GAM. **(b)** Lactation-stage effects for the Protein GAM.

Figure 3: Overview of Lactation Stage Effects.



(a) Country effects for the Energy GAM.

(b) Country effects for the Protein GAM.

Figure 4: Overview of Country Effects.

5.3.3 Overall Feature Effects

These findings demonstrate the intrinsic interpretability and algorithmic fidelity of the GAM architecture as a transparent glass-box system. However, the performance metrics highlight a faithfulness gap; the models currently mirror researcher-defined mathematical equations rather than independently discovered biological determinants, establishing a baseline for future retraining on empirical clinical outcomes.

5.4 Stage-Based Recommendation Evaluation

A controlled experiment was conducted to determine whether changes in lactation-related information propagated through the complete recommendation pipeline and influenced the final nutritional targets and meal recommendations. The same maternal characteristics and food preferences were maintained across the three test cases, while the child's age was varied to represent Early (0–6 months), Mid (7–12 months), and Late (> 12 months) lactation. Because the lactation stage is derived from child age, changing the child's age altered both the continuous child-age input and the corresponding lactation-stage category supplied to the GAM, while the remaining user inputs were held constant.

The three test cases produced different energy requirements. The GAM-predicted energy target was 2,265 kcal/day for Early lactation, 1,977 kcal/day for Mid lactation, and 2,328 kcal/day for Late lactation. The protein target remained 71.5 g/day across the three test cases. The MILP generated a valid meal plan for each stage, with the final recommendations remaining within the permitted energy and protein ranges.

Table 1: Results of the controlled lactation-stage recommendation experiment

Lactation stage	**Energy target (kcal/day)**	**Protein target (g/day)**	**Outcome**
Early	2,265	71.5	Valid plan generated
Mid	1,977	71.5	Valid plan generated
Late	2,328	71.5	Valid plan generated

The ordering of the GAM-predicted energy requirements did not directly reproduce the deterministic stage adjustments used during target construction, where Early, Mid, and Late lactation were assigned +330 kcal/day, +400 kcal/day, and +0 kcal/day, respectively. This difference occurs because the system-level experiment did not vary the categorical lactation-stage feature in isolation. Child age and its derived lactation-stage category were both included as predictors in the trained GAM; therefore, changing child age between the three test cases altered two related model inputs simultaneously. The resulting predictions represent the combined fitted effects of these variables rather than the categorical stage adjustment alone. This finding is consistent with the derived-target limitation discussed in Section 5.3 and highlights the importance of future model retraining and feature validation using directly observed nutritional outcomes.

The resulting meal plans also changed between the three test cases. For example, the Early recommendation included a banana-mango oat breakfast bowl and an egg vegetable rice-noodle supper, whereas the Mid recommendation selected an egg-spinach oatmeal breakfast bowl and a vegetable egg noodle bowl. The Late recommendation returned to a banana-mango oat breakfast but used different quantities and supporting foods, including a larger supper portion.

Micronutrient adequacy was broadly maintained across the three recommendations. The active hard micronutrient targets shown in the reports were satisfied, while Vitamin D remained slightly below its 10 μg/day planning target in all three cases: 9.66 μg in Early, 9.80 μg in Mid, and 8.86 μg in Late. Vitamin D was therefore correctly reported as requiring additional attention rather than preventing generation of the meal plan.

Overall, the controlled experiment demonstrated that lactation-related information propagated beyond the GAM component into the complete recommendation pipeline. Changing child age and its corresponding derived lactation-stage classification resulted in different GAM-derived energy requirements, which subsequently influenced the MILP-generated food selections and portion quantities. Protein remained unchanged across the three cases, consistent with its predominantly weight-based target construction. The experiment therefore demonstrates the functional propagation of lactation-related inputs through the system while also identifying a limitation of the current GAM trained on derived nutritional targets.

5.5 Controlled Maternal-Weight Evaluation

A controlled maternal-weight experiment was conducted to examine whether changes in maternal weight propagated through the GAM and subsequent food-recommendation components. Three test cases were evaluated using maternal weights of 55 kg, 65 kg, and 75 kg, while the remaining maternal characteristics, lactation-related information, and food preferences were held constant.

The predicted protein requirement increased systematically with maternal weight. At 55 kg, the Protein GAM produced a target of 60.5 g/day; at 65 kg, the target increased to 71.5 g/day; and at 75 kg, it increased to 82.5 g/day. These values correspond exactly to the deterministic protein relationship of 1.1 g/kg body weight used during target construction. The energy requirement also increased across the three cases, from 2,121 kcal/day at 55 kg to 2,215 kcal/day at 65 kg and 2,307 kcal/day at 75 kg, consistent with maternal weight also contributing to the Estimated Energy Requirement framework.

Table 2: Results of the controlled maternal-weight recommendation experiment

Maternal weight (kg)	**Energy target (kcal/day)**	**Protein target (g/day)**	**Outcome**
55	2,121	60.5	Valid plan generated
65	2,215	71.5	Valid plan generated
75	2,307	82.5	Valid plan generated

The changes in nutritional requirements also propagated to the MILP recommendation component. Different food combinations and portion quantities were generated across the three weight conditions while the plans remained within the applicable nutritional constraints. This demonstrates that changes in maternal weight affected not only the displayed nutritional targets but also the downstream food recommendations produced by the integrated system.

Overall, the controlled experiment provides system-level confirmation of the strong weight dependence identified during GAM interpretation. In particular, the Protein GAM reproduced the deterministic 1.1 g/kg relationship across all three test conditions, while the MILP adapted the resulting meal plans to the changed nutritional requirements. As with the broader GAM evaluation, this finding demonstrates fidelity to the implemented target-generation framework rather than independent clinical validation.

5.6 Food Recommendation System Evaluation

The food recommendation component was evaluated using a series of functional test scenarios designed to examine nutritional constraint satisfaction, dietary filtering, allergen handling, cuisine preferences, food exclusions, meal structure, and behaviour under increasingly restrictive combinations of user preferences. The tests included unrestricted, vegetarian, vegan, allergy-restricted, cuisine-specific, and excluded-food scenarios. For each successfully generated recommendation, the final validation stage checked whether the plan remained within the permitted energy and protein ranges, satisfied the active hard micronutrient constraints, and complied with the applicable diet, allergy, and excluded-food restrictions.

Table 3: Representative functional test scenarios for the food recommendation system

Scenario	**Main condition tested**	**Outcome**	**Key observation**
Unrestricted diet	Baseline recommendation	Valid plan generated	Energy, protein and active hard micronutrient requirements were satisfied with no dietary restrictions.
Vegetarian	Diet filtering	Valid plan generated	Vegetarian-compatible meals were selected while nutritional constraints remained satisfied.
Vegan	Restrictive diet	Valid plan generated	Hard nutritional constraints remained satisfied; Vitamin B12 and Vitamin D were identified separately where the available plant-based foods could not meet the displayed planning targets.
Vegan + nut allergy	Diet + allergen filtering	Valid plan generated	Conservative nut-allergy filtering removed 1,789 food records while a valid plan was still produced.
Vegan + soy allergy	Highly restrictive diet + allergen filtering	Valid plan generated	Soy filtering removed 1,743 records; the system retained diet/allergen safety and produced a validated plan.
Omnivore + African cuisine + chicken exclusion	Cuisine + food exclusion	Valid plan generated	Chicken-containing foods and recipes were excluded and suitable African prepared dishes were prioritised.
Omnivore + North American cuisine + egg exclusion	Cuisine + food exclusion	Valid plan generated	Egg-containing options were excluded while the remaining candidate pool supported a valid recommendation.
Vegan + Latin American cuisine + rice exclusion	Diet + cuisine + exclusion	Valid plan generated using fallback	The strict curated-main configuration was infeasible, but a valid plan was obtained after relaxing secondary meal-variety constraints while retaining nutritional and safety constraints.
Vegetarian + lactose-free + Mediterranean cuisine	Multiple simultaneous restrictions	Valid plan generated using fallback	The system retained the selected dietary restrictions and conservative lactose-free filtering while relaxing secondary meal-variety constraints when required.
Vegan + Indian cuisine	Diet + cuisine	Valid plan generated	Cuisine-compatible curated recipes were prioritised and Vitamin B12 and Vitamin D were appropriately reported as requiring additional attention.

The evaluation demonstrated that the recommendation pipeline responded to changes in user preferences rather than applying the same candidate pool to every user. For example, the unrestricted test retained 4,981 of 4,990 catalogue rows after preference filtering, whereas restrictive vegan and allergen scenarios substantially reduced the available food pool. In the vegan nut-allergy scenario, conservative allergen filtering removed 1,789 foods, while the vegan soy-allergy scenario removed 1,743 foods. Despite these reductions, both scenarios produced recommendations that passed the final energy, protein, active hard micronutrient, diet, allergy and exclusion checks.

The tests also confirmed that food exclusions were applied beyond the displayed food name. Recipe ingredients and ingredient keys were considered during filtering, allowing exclusions to affect prepared dishes as well as individual foods. In the vegan Latin American scenario with rice excluded, 305 foods or recipes were removed by the excluded-food filter. The resulting recommendation remained within the energy and protein ranges and satisfied the active hard micronutrient constraints. Similarly, exclusion of chicken under an African cuisine preference and exclusion of eggs under a North American cuisine preference produced validated plans without the excluded foods.

Cuisine filtering was also successfully incorporated without preventing general foods such as fruit, vegetables and grains from being used as supporting meal components. In several scenarios, curated cuisine-specific recipes were successfully prioritised for the main meals. Where the stricter curated-main configuration could not produce a feasible solution, the system allowed suitable USDA prepared dishes as a fallback rather than relaxing the core nutritional or dietary-safety requirements. [20] This behaviour was observed under more restrictive combinations such as vegan Latin American with rice excluded and vegetarian lactose-free Mediterranean preferences.

The vegan tests additionally demonstrated the system's handling of nutrients for which the eligible food catalogue could not reliably support hard optimisation constraints. Vitamin B12 and Vitamin D were retained as displayed planning targets and were explicitly reported as requiring additional attention when the generated food plan did not meet them. [21] For example, the vegan nut-allergy plan provided 0.09 μg Vitamin B12 and 0.00 μg Vitamin D against planning targets of 3.80 μg and 10.00 μg, respectively, while the remaining active hard constraints were satisfied. This approach allows the system to maintain mathematical feasibility while transparently identifying nutrients that were not adequately supplied by the generated meal plan.

Overall, the functional evaluation indicates that the MILP recommendation component can generate nutritionally constrained meal plans across a range of dietary and preference configurations while enforcing the principal safety filters and explicitly reporting limitations when particular nutrient targets cannot reliably be satisfied from the eligible food catalogue. More restrictive scenarios reduced the feasible candidate pool and occasionally required relaxation of secondary meal-variety constraints; however, the reported fallback cases retained the core energy, protein, active hard micronutrient, diet, allergy, exclusion, meal-structure and portion constraints. These results therefore support the functional correctness of the recommendation pipeline across the tested scenarios, rather than establishing clinical effectiveness or generalising performance to all possible user configurations.

6. Discussion

6.1 Interpretation of Results and Algorithmic Fidelity

The results confirm that Generalised Additive Models (GAMs) provide a robust glass-box architecture for maternal health informatics. The Energy GAM ($R^2 = 0.9722$) demonstrated high overall fidelity to Estimated Energy Requirement (EER) targets across weight, height, age, and stage variables, while the Protein GAM ($R^2 = 1.0$) accurately reproduced the deterministic $1.1g/kg formula$. The absence of country-specific variation across both models confirms that they remained mathematically faithful to the target-generation logic, which did not encode population-level variables.

6.2 Interpretability and Personalisation

By replacing opaque black-box algorithms with additive smooth functions and partial-effect plots, the system provides intrinsic interpretability. Visualizing how individual maternal attributes drive predictions offers the actionable recourse and transparent reasoning necessary for clinical decision-support, healthcare practitioner auditing, and maternal engagement.

6.3 Derived Target Limitation and Future Work

The primary limitation of this study is its reliance on derived deterministic targets rather than directly observed individual-level requirements. Consequently, the high performance metrics highlight a faithfulness gap. The models currently mirror researcher-specified mathematical relationships rather than independently discovered biological truths.

This framework serves as a validated proof-of-concept architecture. Future research must focus on retraining these GAMs on empirical datasets containing directly measured individual nutritional outcomes to evaluate real-world predictive validity, clinical efficacy, and generalizability across diverse populations.

6.4 Integration With Mixed-Integer Linear Programming

The GAM and reference-based micronutrient modules provide complementary nutritional targets to the food recommendation model. Maternal characteristics and lactation stage are processed by the trained GAMs to generate personalised daily energy and protein requirements. In parallel, the micronutrient module determines the applicable vitamin and mineral reference requirements, including energy-dependent and age-dependent requirements where relevant.

These outputs are combined to form the nutritional targets supplied to the Mixed-Integer Linear Programming (MILP) model. The GAM-predicted energy and protein values define the permitted daily intake ranges, while eligible micronutrient requirements are incorporated as minimum constraints. The optimiser then evaluates the filtered food and recipe candidates and determines suitable food selections and portion sizes that satisfy these requirements together with the user's dietary restrictions and meal-structure constraints.

6.5 Evaluation of the Food Recommendation System

The functional evaluation demonstrated that the recommendation system could translate the GAM-derived energy and protein requirements and reference-based micronutrient targets into valid meal plans across a range of dietary and preference configurations. Successful recommendations were generated for unrestricted, vegetarian and vegan diets, as well as scenarios involving allergies, cuisine preferences and user-specified food exclusions. The generated plans remained within the permitted energy and protein ranges and satisfied the active hard micronutrient constraints, while final validation ensured compliance with the applicable dietary, allergy and excluded-food restrictions.

The controlled lactation-stage experiment further demonstrated that stage information propagated beyond the GAM component into the final recommendation pipeline. When the maternal characteristics and food preferences were held constant and child age was varied across the Early, Mid, and Late lactation categories, both the continuous child-age input and its corresponding derived lactation-stage classification changed. The GAM-derived energy requirement subsequently changed, and the MILP selected different food combinations and portion quantities. Protein remained unchanged across the three controlled cases, indicating that not all predicted nutritional requirements were equally sensitive to the stage change. These findings show that the lactation stage can influence the final recommendation through its effect on the model-derived nutritional targets rather than functioning only as a descriptive characteristic displayed by the web interface.

The controlled maternal-weight experiment provided further evidence that changes in model inputs propagated through the integrated recommendation pipeline. Increasing maternal weight from 55 kg to 75 kg produced systematic increases in both the GAM-derived energy and protein requirements, with the protein estimates reproducing the deterministic 1.1 g/kg relationship used during target construction. These changes subsequently resulted in different food combinations and portion quantities generated by the MILP. This experiment therefore supports the functional integration of the GAM and optimisation components while, consistent with the derived-target limitation, demonstrating fidelity to the implemented nutritional logic rather than independent clinical validity.

The more restrictive test scenarios also highlighted limitations arising from food-based optimisation and the available food catalogue. Vegan plans could satisfy the active hard nutritional constraints but did not reliably meet the displayed Vitamin B12 and Vitamin D planning targets using the available eligible foods. Rather than treating these nutrients as satisfied, the system reported them as requiring additional attention. Similarly, some highly restrictive combinations required relaxation of secondary meal-variety constraints before a feasible solution could be obtained, while the core nutritional, dietary-safety, allergy, exclusion, meal-structure and portion constraints remained enforced. These findings demonstrate the importance of distinguishing between nutritional feasibility, practical meal preferences and limitations arising from the available food composition catalogue.

Overall, the evaluation supports the functional correctness of the integrated recommendation pipeline across the tested scenarios. However, the results represent computational and functional evaluation rather than evidence of clinical effectiveness. Further evaluation using real users, dietitian assessment and observed dietary or health outcomes would be required to determine the practical acceptability and clinical value of the generated recommendations.

6.6 Comparison With Existing Research

The developed system combines elements that have largely been investigated separately in previous research. The NUTRISHIELD study [1] similarly demonstrated the value of personalised nutrition guidance for lactating mothers through a digital platform. However, NUTRISHIELD incorporated detailed clinical, dietary and

biological information, including human milk and laboratory data. In contrast, the present system uses routinely available maternal characteristics and lactation stage to estimate personalised energy and protein requirements, reducing the dependence on specialised biological measurements while maintaining an interpretable recommendation process.

The Linear Programming approach used for Chinese urban lactating women [8] is more closely related to the food recommendation component of this study. Both approaches use optimisation to translate nutritional requirements into food-based recommendations and consider the challenge of satisfying micronutrient requirements during lactation. However, the previous study focused on developing optimised food-based dietary guidelines, whereas the present system integrates optimisation with GAM-derived personalised energy and protein requirements, reference-based micronutrient targets, user dietary restrictions, and a web-based recommendation interface.

The use of GAMs also reflects previous research demonstrating their value as interpretable models in health-related applications [12]. Similar to this research, the present study uses GAMs to examine how predictor variables influence model outputs while retaining greater transparency than conventional black-box approaches. The main difference lies in the application domain: rather than predicting obesity-related health outcomes, the GAMs in this study estimate personalised energy and protein requirements for lactating women and provide these estimates to the subsequent optimisation stage.

Overall, the contribution of the present study lies in the integration of these approaches rather than in treating personalised nutrition, interpretable modelling, and dietary optimisation as separate problems. The resulting framework connects maternal characteristics and lactation stage with interpretable GAM predictions, reference-based micronutrient requirements, MILP-based meal optimisation, validation, and a user-facing recommendation system. The functional evaluation further demonstrated that these components operated as an integrated pipeline across the tested dietary and preference scenarios. This addresses the integration gap identified in the reviewed literature while providing a computational foundation for further real-world evaluation.

Table 4: Feature-based comparison of the present study with related approaches

Feature	**NUTRI-SHIELD [1]**	**Yu et al. [8]**	**Awe et al. [12]**	**Present Study**
Lactating-women focus	✓	✓	—	✓
Personalised/individualised approach	✓	✓	—	✓
Lactation stage/duration considered	✓	✓	—	✓
Interpretable GAM modelling	—	—	✓	✓
Mathematical dietary optimisation	—	✓	—	✓
Micronutrient information/requirements	✓	✓	—	✓
Individual dietary preferences/restrictions	✓	∘	—	✓
Food/diet recommendations	✓	✓	—	✓
Digital/user-facing nutrition platform	✓	—	—	✓
GAM + dietary optimisation integration	—	—	—	✓

✓ Included
∘ Partially/indirectly included
— Not included/reported

7. Conclusion and Future Work

This study successfully established a stage-aware, intrinsically interpretable framework for estimating the personalised nutritional requirements of lactating mothers and translating them into actionable meal plans. Utilising a validated DHS dataset of 12,176 observations across five African nations, separate Generalised Additive Models (GAMs) were developed, achieving high fidelity to constructed target equations (R^2=0.9722 for energy; R^2=1.0 for protein). These macronutrient estimates were combined with a reference-based micronutrient module and integrated into a Mixed-Integer Linear Programming (MILP) optimisation engine. The resulting pipeline generates nutritionally compliant daily food recommendations while respecting user preferences, allergies, dietary

constraints, portion limits, and meal structures via a web interface.

The primary limitation remains the reliance on researcher-derived deterministic targets, meaning high GAM performance reflects algorithmic fidelity to implemented logic rather than clinical predictive validity. Nevertheless, this research demonstrates the feasibility of combining interpretable machine learning, reference micronutrient benchmarks, and mathematical optimisation in a transparent digital health intervention. Future research must focus on evaluating the framework against empirical, individual-level clinical outcomes to establish real-world predictive validity and generalizability across diverse maternal populations.

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9. Declaration

Administrative and Ethics

This research received no external funding, and the authors declare no competing interests. Ethics approval, participant consent, and consent for publication were not applicable, as the study exclusively utilized publicly available secondary Demographic and Health Survey (DHS) data without direct human contact.

Data, Materials and Code Availability

Demographic inputs were obtained from public DHS datasets (Burundi, Ethiopia, Kenya, Nigeria, South Africa), and food composition data were drawn from USDA FoodData Central. All processed datasets, model artefacts, curated recipe catalogues, and source code (covering GAM modelling, MILP optimisation, and the web application) are available from the authors upon reasonable request, subject to provider data-use restrictions.

Author Contributions

Sasthi Naidoo: Developed the web application, reference-based micronutrient component, food and recipe data pipeline, MILP optimisation engine, UI, and functional recommendation testing.

Varushka Dayal: Developed and evaluated the Generalised Additive Models (GAMs), including dataset preparation, target variable generation, model training, predictive performance evaluation, and interpretability analysis.

Both Authors: Contributed jointly to the overall research design, system integration, interpretation of results, and final manuscript preparation/review.

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Appendix

Appendix A: Full List of Dataset Variables

Variable	DHS Variable	Description	Unit/ Coding	Used in GAM?
Age	V012	Respondent's age	Years	Yes
Weight	V437	Respondent's weight	kg	Yes
Height	V438	Respondent's height	cm	Yes
BMI	V445	Body Mass Index	kg/m^2	No
Currently breastfeeding	V404	If respondent is currently breastfeeding	Binary	Selection Criterion
Currently Pregnant	V213	If respondent is currently pregnant	Binary	Yes
Number of living children	V218	Number of living children	Count	Yes
Child Age	B19	Current age of child	Months	Yes
Anemia status	V457	Maternal Anemia Classification	Categorical	No
Lactation Stage	Derived	Based on child's age	Mid/Early/ Late	Yes
Country	Derived	DHS survey country	Categorical	Yes
Y_Energy	Derived	Individualised energy requirement	kcal/day	Target
Y_Protein	Derived	Individualised protein requirement	g/day	Target

Table 5: Full list of dataset variables used in the study.

Appendix B: Nutritional Equations Used for Target Generation

B.1 Protein Requirement

The protein requirement was calculated using:

$$\text{Protein Requirement} = 1.1 \times \text{Body Weight}$$

[22]

where:

- Protein requirement is expressed in g/day.
- Maternal body weight is expressed in kg.
- The resulting value is an individualized continuous target.

B.2 Estimated Energy Requirement

The Estimated Energy Requirement (EER) was calculated using:

$$EER = 354 - (6.91 \times \text{Age}) + PA\,[(9.36 \times \text{Weight}) + (726 \times \text{Height}_m)]$$

[23]

where:

- Age: maternal age in years.
- Weight: maternal weight in kg.
- Height_m: maternal height in metres.
- PA: physical activity coefficient.
- PA = 1 in this study.

B.3 Lactation Adjustment

Lactation Stage	Energy Adjustment
Early (0–6 months)	+330 kcal/day
Mid (7–12 months)	+400 kcal/day
Late (> 12 months)	+0 kcal/day

Table 6: Energy adjustments applied according to lactation stage.

B.4 Minimum Energy Threshold

The final energy target was calculated as:

$$\text{Final Energy Target} = \max\left(\text{Calculated Energy Requirement}, 1800 \text{ kcal/day}\right)$$

Appendix C: GAM Hyperparameters and Training Configuration

Parameter	Protein GAM	Energy GAM
Model	LinearGAM	LinearGAM
Input Variables	14 encoded features	14 encoded features
Target	Y_Protein	Y_Energy
Train/Test Split	80/20	80/20
Random State	42	42
Categorical Encoding	One-hot	One-hot
Lactation Encoding	One-hot, all levels retained	One-hot, all levels retained
Country Encoding	One-hot, all levels retained	One-hot, all levels retained
Evaluation Metrics	MAE, RMSE, R^2	MAE, RMSE, R^2

Table 7: GAM hyperparameters and training configuration.

Appendix D: Micronutrient Reference Requirements

The micronutrient requirements used in the food recommendation module were obtained from the Health Council of the Netherlands [18], *Dietary Reference Values for Vitamins and Minerals for Lactating Women*. These values are used as reference-based nutritional planning targets within the Mixed-Integer Linear Programming (MILP) model.

Nutrient	Requirement	Unit	Reference Type	Condition	Implementation in Optimisation
Vitamin A	1100	μg RAE/day	AI	Lactating women	Hard minimum*
Vitamin B1	0.1 × energy (MJ) + 0.16	mg/day	PRI	Energy-dependent	Hard minimum*
Vitamin B2	2	mg/day	PRI	Lactating women	Hard minimum*
Vitamin B6	1.7	mg/day	PRI	Lactating women	Hard minimum*
Folate	500	μg DFE/day	PRI	Lactating women	Hard minimum*
Vitamin B12	3.8	μg/day	PRI	Lactating women	Hard minimum*; report-only for vegan plans
Vitamin C	135	mg/day	PRI	Lactating women	Hard minimum*
Vitamin D	10	μg/day	AI	Lactating women	Soft planning target
Calcium	1000	mg/day	PRI	Age 18–24	Hard minimum*
Calcium	950	mg/day	PRI	Age $\geq$ 25	Hard minimum*
Iron	16	mg/day	PRI	Lactating women	Hard minimum*
Magnesium	300	mg/day	AI	Lactating women	Hard minimum*
Selenium	85	μg/day	AI	Lactating women	Hard minimum*
Zinc	9.9	mg/day	PRI	Lactating women	Hard minimum*

Table 8: Micronutrient reference requirements used by the system.

Note: Hard micronutrient constraints are activated only when at least three eligible foods contain positive recorded values for the nutrient. Vitamin B1 is calculated from the personalised energy requirement. Vitamin D is a soft planning target, while Vitamin B12 is report-only for vegan plans. Iodine is excluded because of insufficient food-composition coverage. Reference values are nutritional planning targets, not individual clinical assessments.

Appendix E: MILP Mathematical Formulation

For each eligible food or recipe i, let $x_i \geq 0$ represent its selected quantity in grams and $y_i \in \{0, 1\}$ indicate whether the item is selected.

The total contribution of nutrient n is:

$$N_n(x) = \sum_{i=1}^{m} \frac{x_i}{100} c_{in} \tag{8}$$

where c_{in} is the nutrient content per 100 g of item i.

The principal optimisation constraints are:

$$0.98E_{\text{GAM}} \leq E(x) \leq 1.02E_{\text{GAM}}, \tag{9}$$

$$0.95P_{\text{GAM}} \leq P(x) \leq 1.05P_{\text{GAM}}, \tag{10}$$

$$N_n(x) \geq R_n, \quad n \in H, \tag{11}$$

$$L_i y_i \leq x_i \leq U_i y_i, \tag{12}$$

$$x_i \geq 0, \quad y_i \in \{0, 1\}. \tag{13}$$

Here, E_{GAM} and P_{GAM} are the GAM-derived requirements; R_n is the reference requirement for an active hard micronutrient $n \in H$; and L_i and U_i are the permitted portion bounds.

The meal structure requires one main selection for breakfast, lunch, snack, and supper, with an optional dessert. For the corresponding eligible main-item sets $\mathcal{B}$, $\mathcal{L}$, $\mathcal{S}_n$, $\mathcal{S}_u$, and $\mathcal{D}$:

$$\sum_{i \in \mathcal{B}} y_i = 1, \qquad \sum_{i \in \mathcal{L}} y_i = 1, \tag{14}$$

$$\sum_{i \in \mathcal{S}_n} y_i = 1, \qquad \sum_{i \in \mathcal{S}_u} y_i = 1, \tag{15}$$

$$\sum_{i \in \mathcal{D}} y_i \leq 1. \tag{16}$$

Additional supporting-food and variety constraints regulate practical meal composition. Dietary type, allergies, cuisine preferences and excluded foods are applied through candidate filtering before optimisation.

The objective function is represented by the following general form:

$$\begin{aligned} \min Z(x, y) = Z_{\text{nutrition}} + Z_{\text{practicality}} \\ + Z_{\text{food quality}} - Z_{\text{preferences}}. \end{aligned} \tag{17}$$

The objective incorporates soft penalties and preferences relating to nutritional balance, food suitability, portion practicality and curated-recipe selection. The MILP is solved using CBC through PuLP, and generated recommendations undergo final nutritional and dietary validation.

Appendix F: Fair Use of AI

We, Sasthi Naidoo and Varushka Dayal, made use of generative AI tools in this report in the following areas: Writing and language refinement (improve clarity, grammar, and structure of the report), LaTeX formatting (assisted with resolving formatting and compilation issues), and programming and technical assistance (assisted with troubleshooting code.)

In all instances, the content was fully interrogated for applicability, accuracy, and plagiarism. Citations were added to statements where applicable to appropriately assign credit to the original authors.